

CS 171 / EE 142: Intro to ML and DM

Christian Shelton

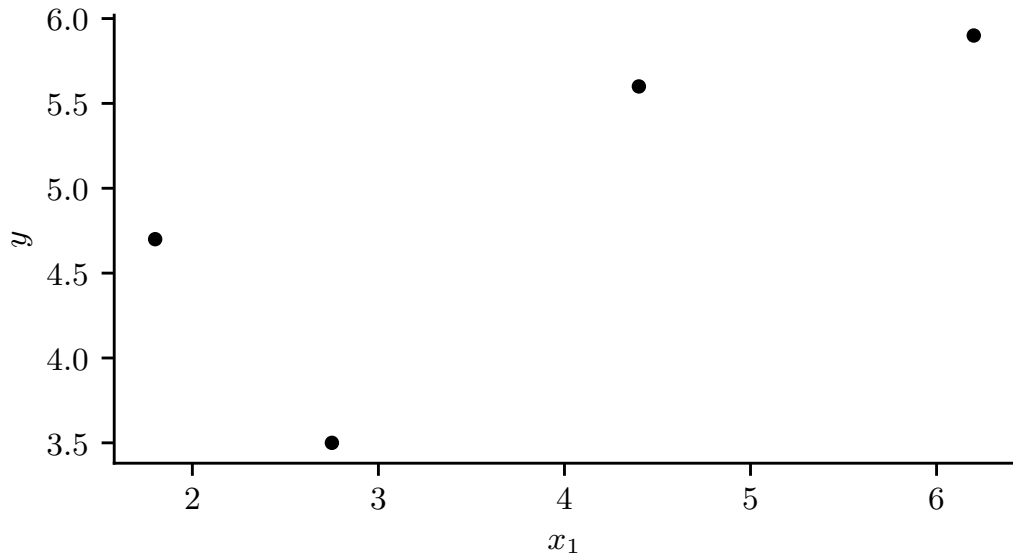
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Slide Set 4: Linear Regression

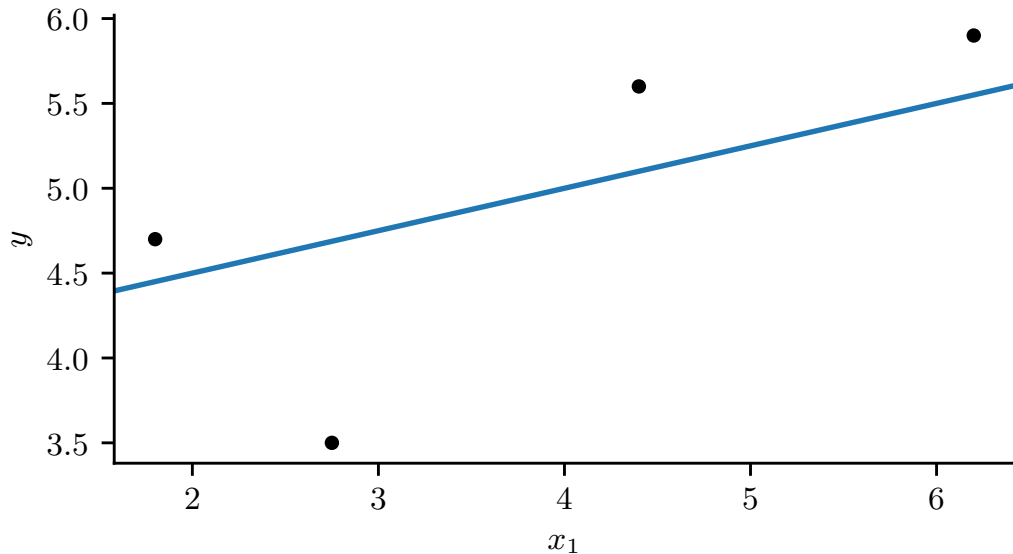


- From UC Riverside
 - ▶ CS 171 / EE 142: Introduction to Machine Learning and Data Mining
 - ▶ Professor Christian Shelton
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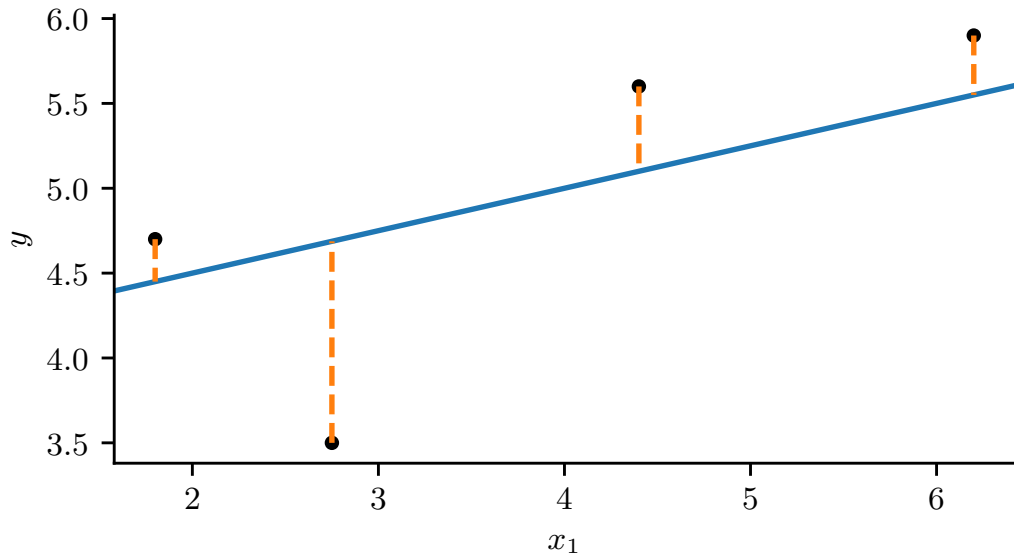
Linear Regression, 1D



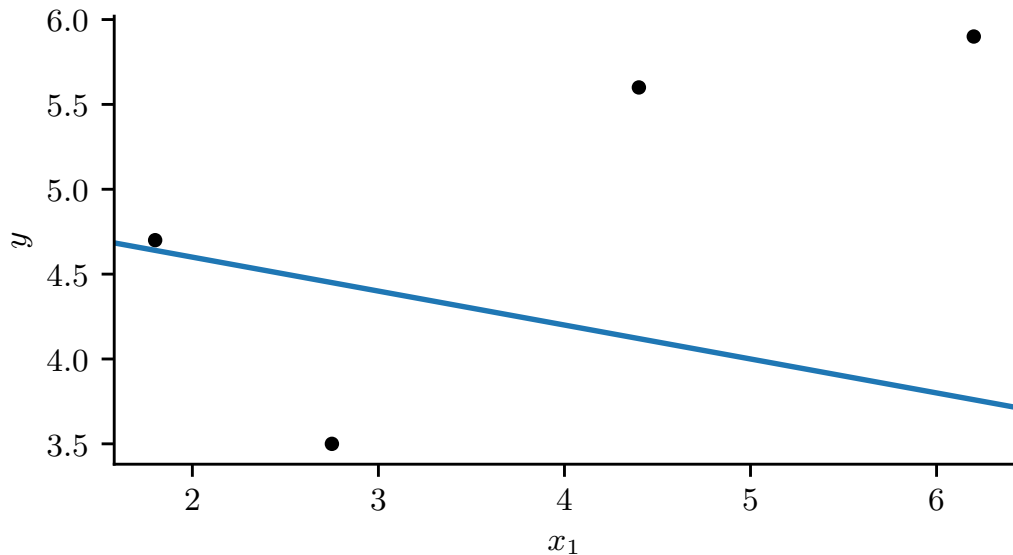
Linear Regression, 1D



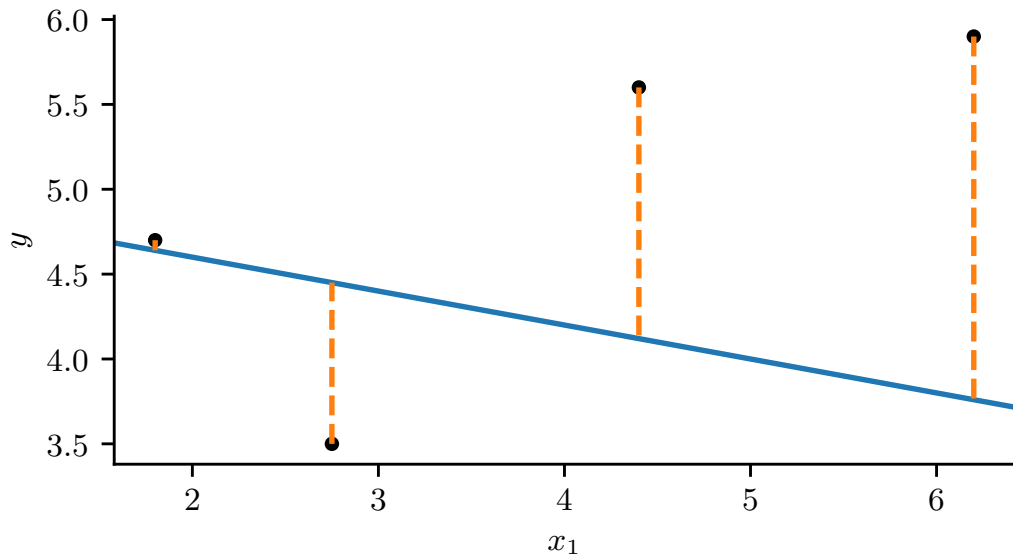
Linear Regression, 1D



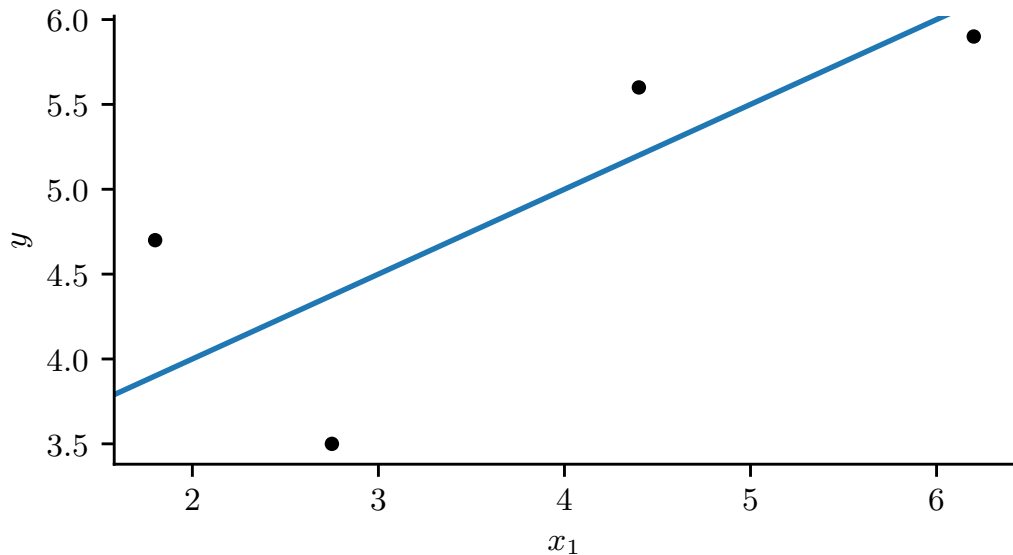
Linear Regression, 1D



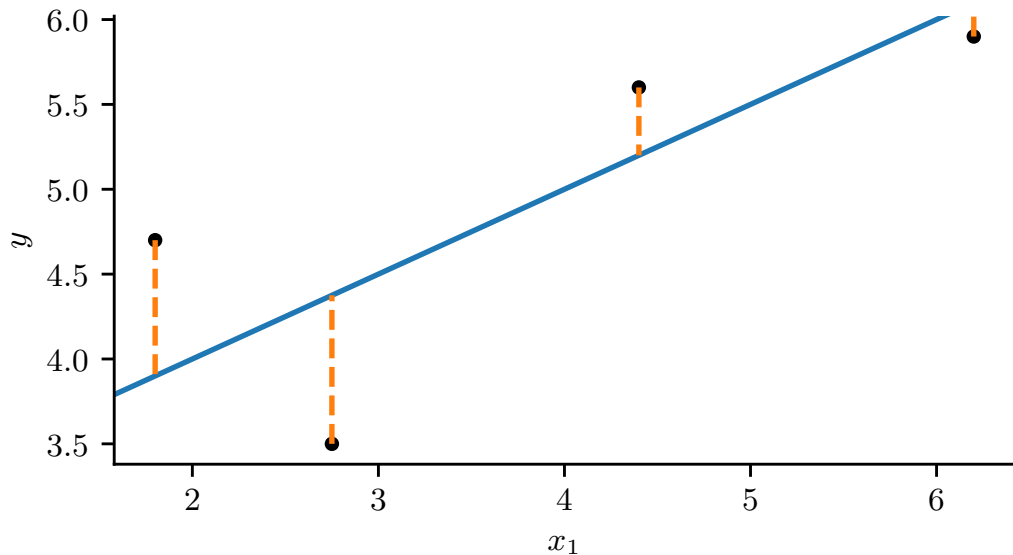
Linear Regression, 1D



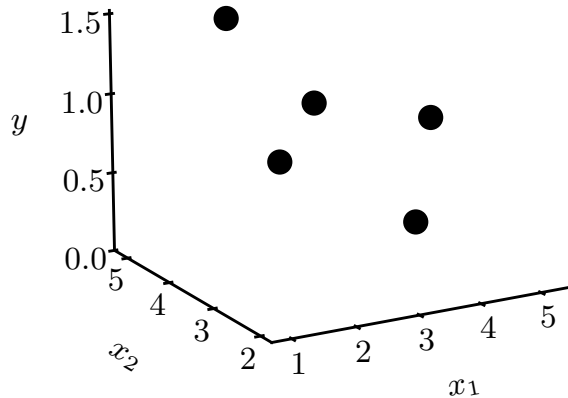
Linear Regression, 1D



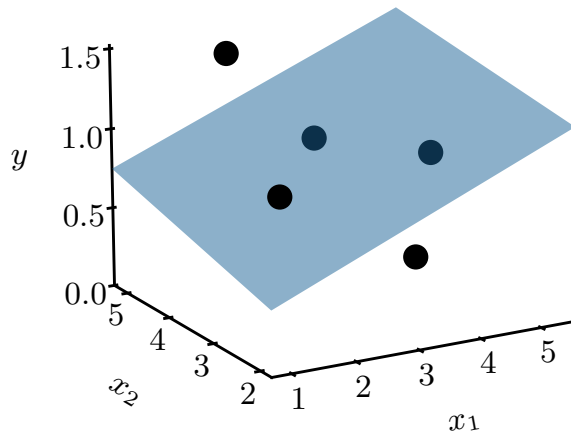
Linear Regression, 1D



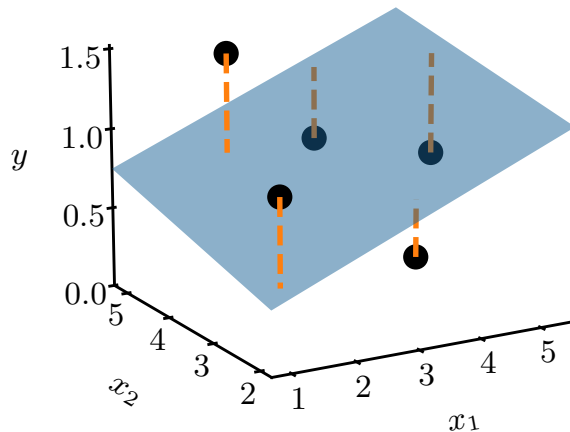
Linear Regression, 2D



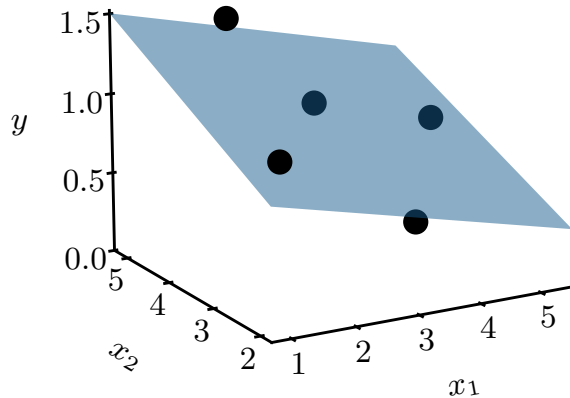
Linear Regression, 2D



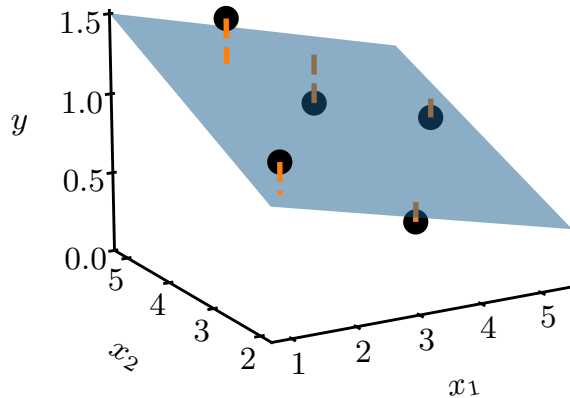
Linear Regression, 2D



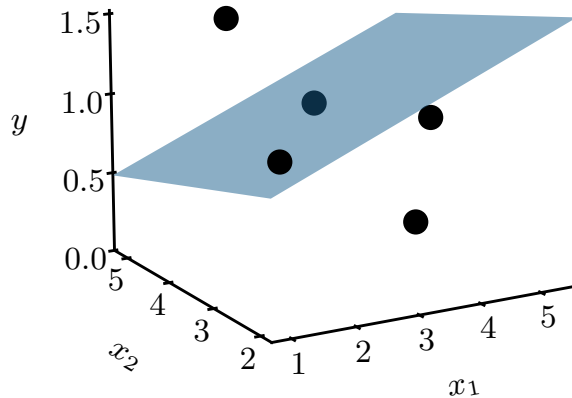
Linear Regression, 2D



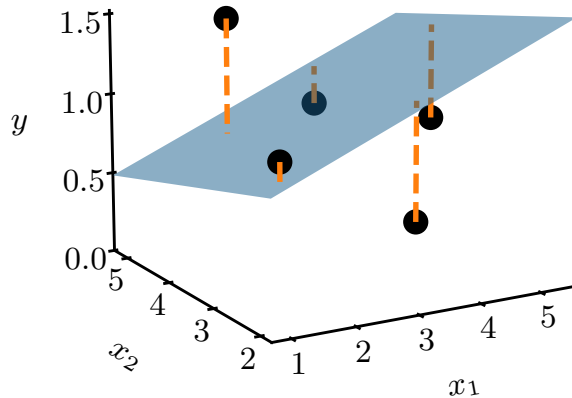
Linear Regression, 2D



Linear Regression, 2D

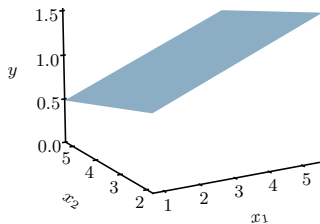


Linear Regression, 2D



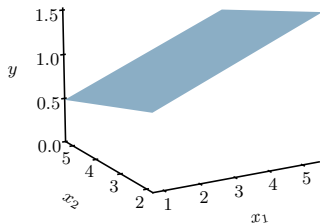
Linear Function Parameterization

$$f(\vec{x}) = b + w_1x_1 + w_2x_2$$



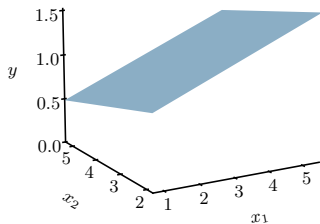
Linear Function Parameterization

$$f(\vec{x}) = \underbrace{w_0}_{\text{aka "b"}} \overbrace{x_0}^{=1} + w_1 x_1 + w_2 x_2$$

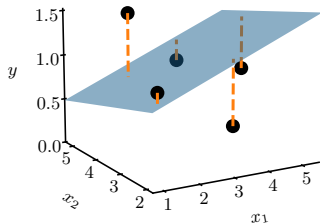


Linear Function Parameterization

$$f(\vec{x}) = \sum_{j=0}^n w_j x_j = \vec{w}^\top \vec{x} = \vec{x}^\top \vec{w}$$



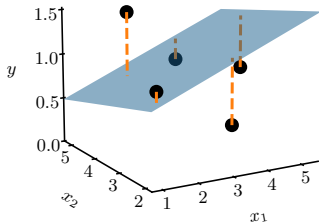
Squared Loss



$$f(\vec{x}) = \sum_{j=0}^n w_j x_j = \vec{w}^\top \vec{x} = \vec{x}^\top \vec{w}$$

$$\underbrace{l(y, \hat{y})}_{\text{per-item loss}} = (y - \hat{y})^2$$

Squared Loss

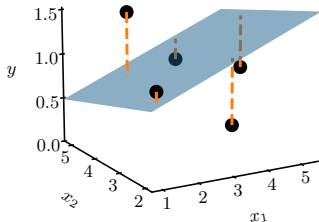


$$f(\vec{x}) = \sum_{j=0}^n w_j x_j = \vec{w}^\top \vec{x} = \vec{x}^\top \vec{w}$$

$$\underbrace{l(y, \hat{y})}_{\text{per-item loss}} = (y - \hat{y})^2$$

$$\underbrace{L}_{\text{training loss}} = \sum_{i=1}^m l(y_i, f(\vec{x}_i))$$

Squared Loss

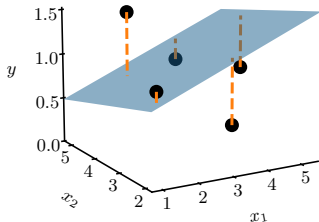


$$f(\vec{x}) = \sum_{j=0}^n w_j x_j = \vec{w}^\top \vec{x} = \vec{x}^\top \vec{w}$$

$$\underbrace{l(y, \hat{y})}_{\text{per-item loss}} = (y - \hat{y})^2$$

$$\underbrace{L}_{\text{training loss}} = \sum_{i=1}^m (y_i - f(\vec{x}_i))^2$$

Squared Loss

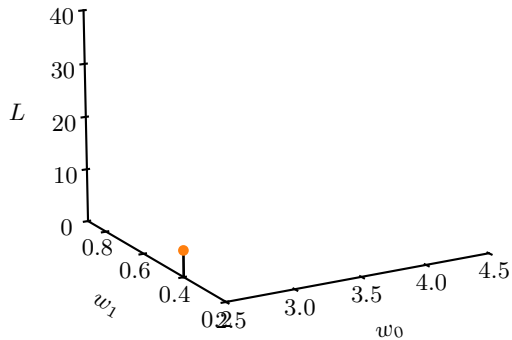
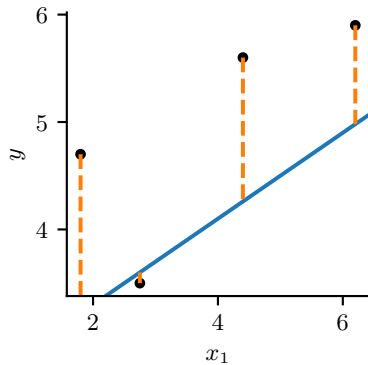


$$f(\vec{x}) = \sum_{j=0}^n w_j x_j = \vec{w}^\top \vec{x} = \vec{x}^\top \vec{w}$$

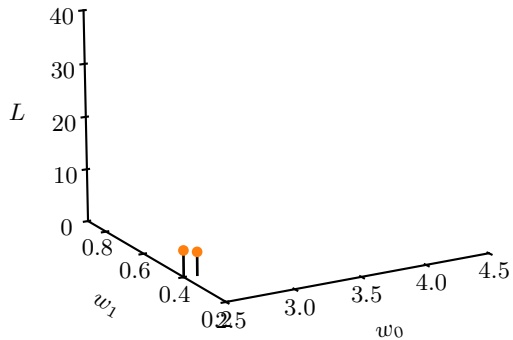
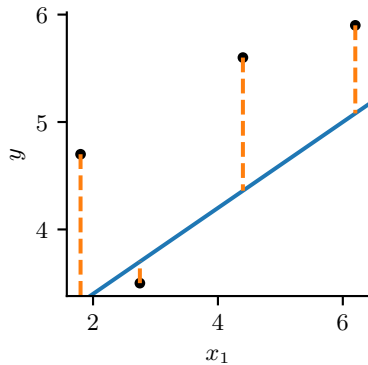
$$\underbrace{l(y, \hat{y})}_{\text{per-item loss}} = (y - \hat{y})^2$$

$$\underbrace{L}_{\text{training loss}} = \sum_{i=1}^m (y_i - f(\vec{x}_i))^2$$
$$= \sum_{i=1}^m \left(y_i - \sum_{j=0}^n w_j x_{i,j} \right)^2$$

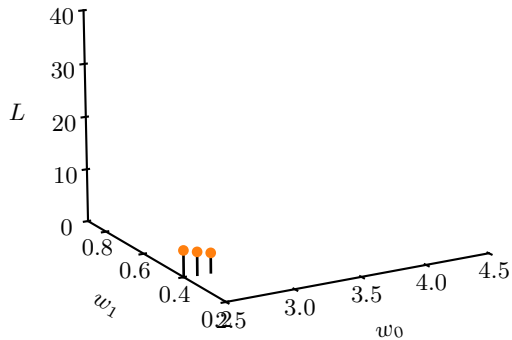
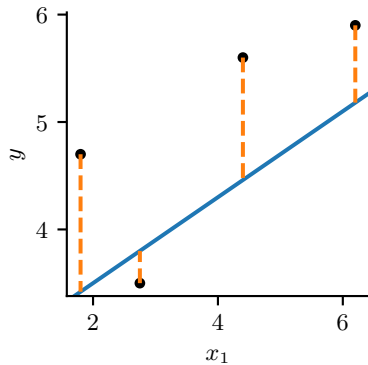
Minimization of Squared Loss



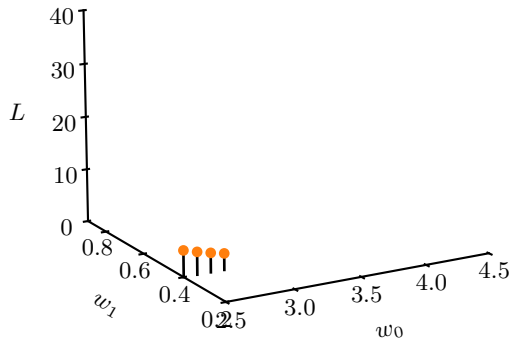
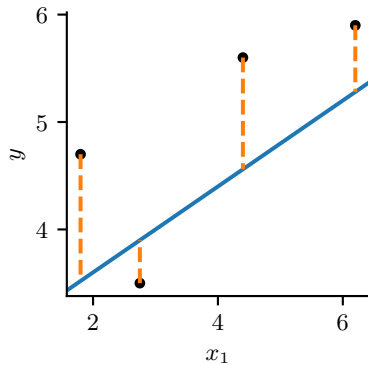
Minimization of Squared Loss



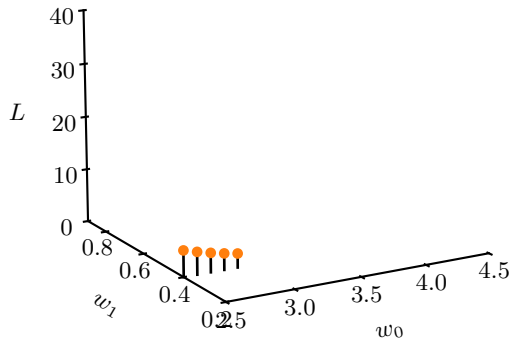
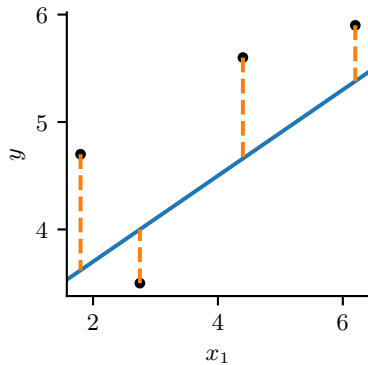
Minimization of Squared Loss



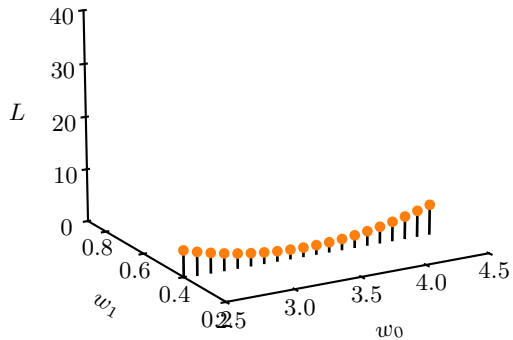
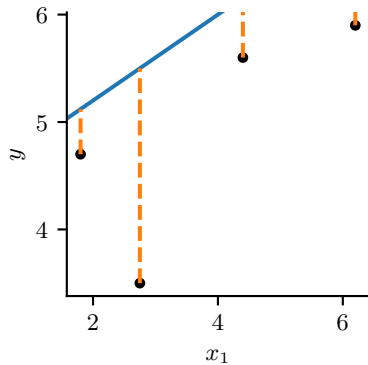
Minimization of Squared Loss



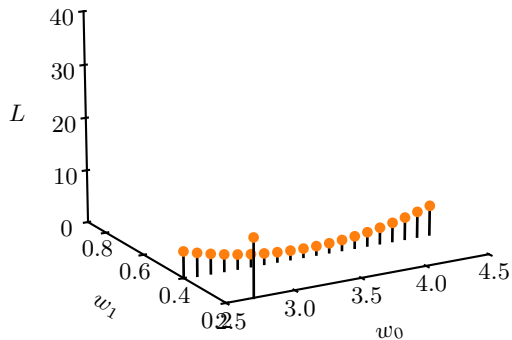
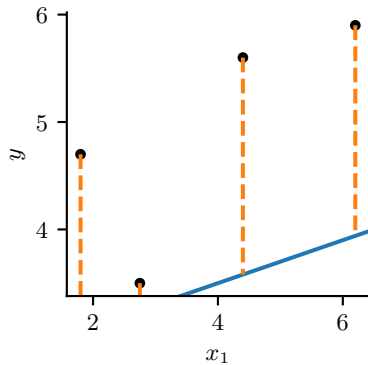
Minimization of Squared Loss



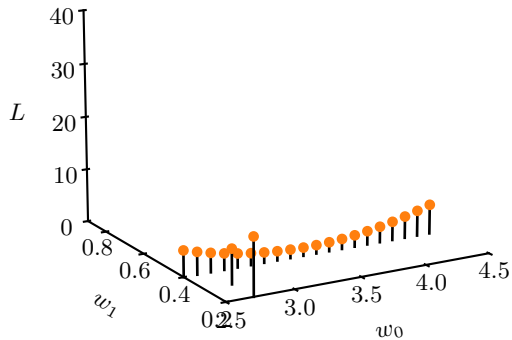
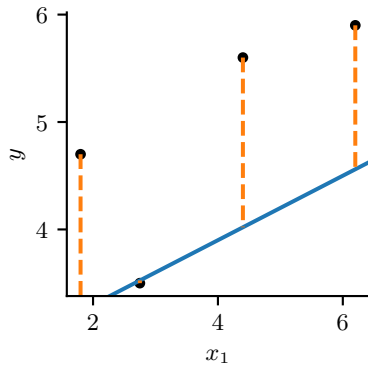
Minimization of Squared Loss



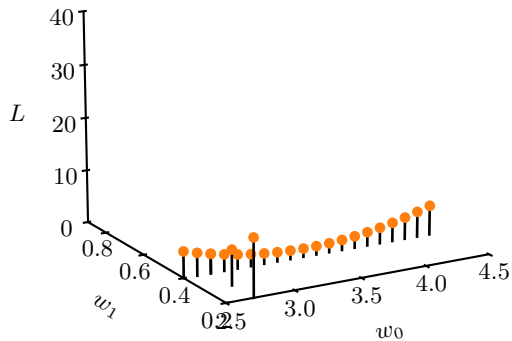
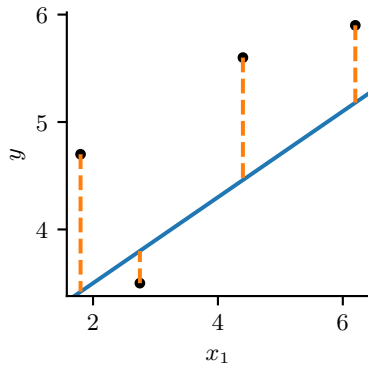
Minimization of Squared Loss



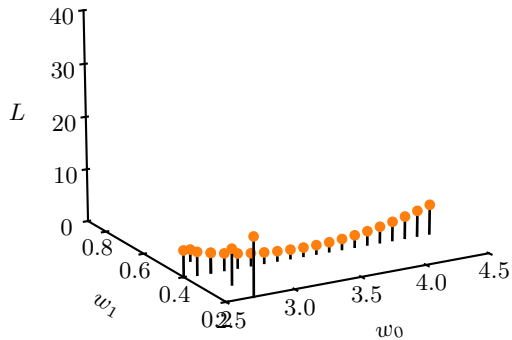
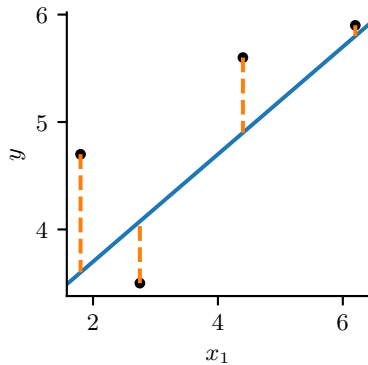
Minimization of Squared Loss



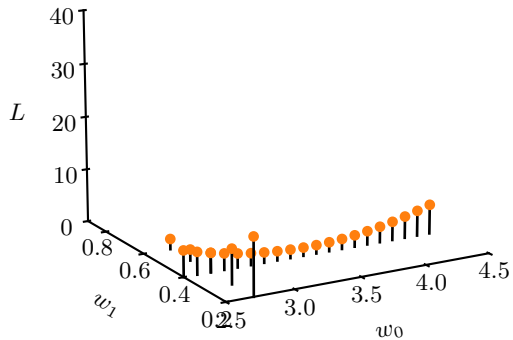
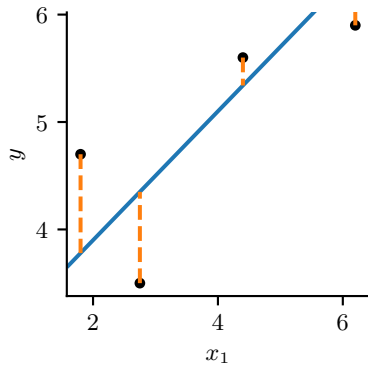
Minimization of Squared Loss



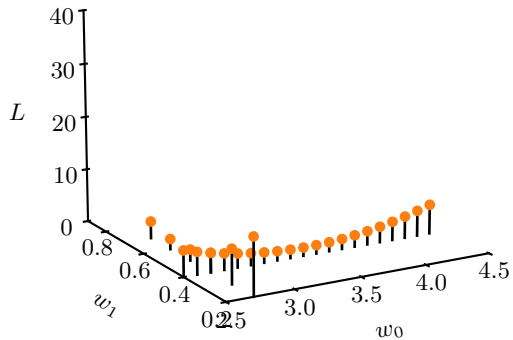
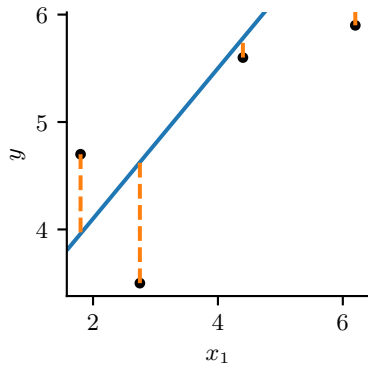
Minimization of Squared Loss



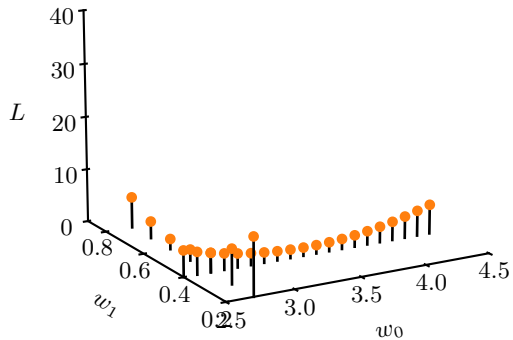
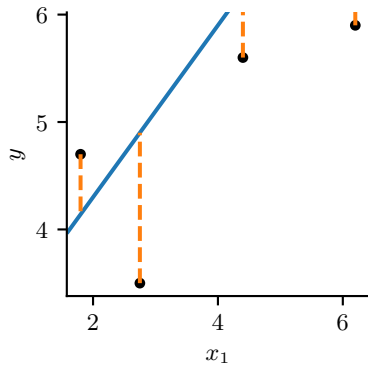
Minimization of Squared Loss



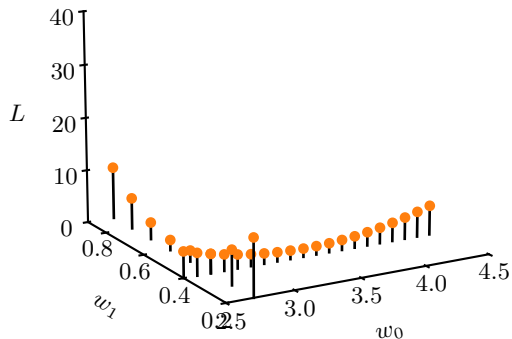
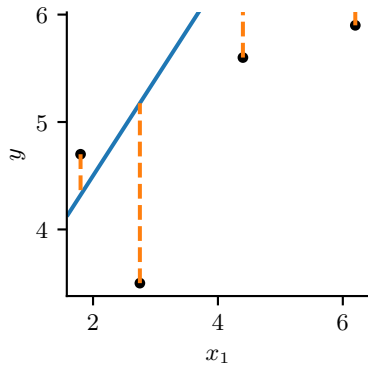
Minimization of Squared Loss



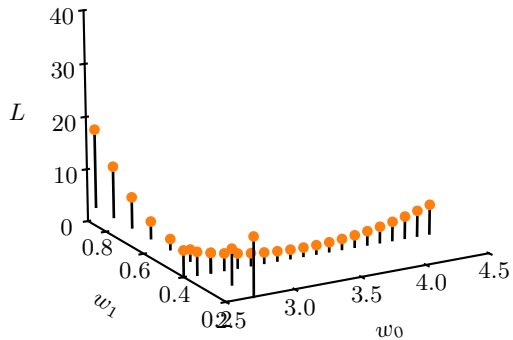
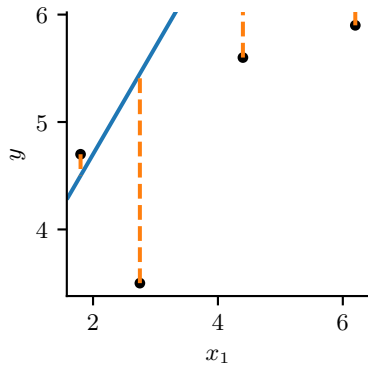
Minimization of Squared Loss



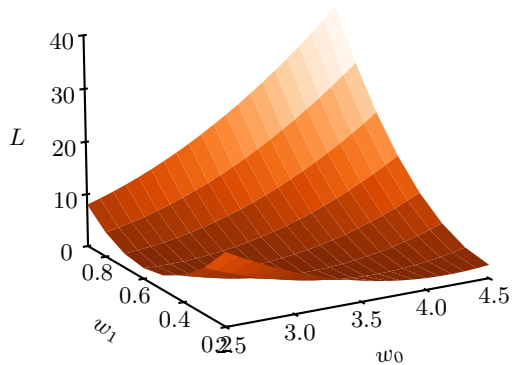
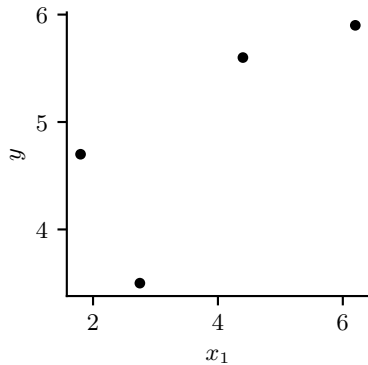
Minimization of Squared Loss



Minimization of Squared Loss



Minimization of Squared Loss



Minimization of Squared Loss

$$L = \sum_{i=1}^m \left(y_i - \sum_{j=0}^n w_j x_{i,j} \right)^2$$

Minimization of Squared Loss

$$L = \sum_{i=1}^m \left(y_i - \sum_{j=0}^n w_j x_{i,j} \right)^2$$

$$\frac{\partial L}{\partial w_0} = \sum_{i=1}^m -2 \left(y_i - \sum_{j=0}^n w_j x_{i,j} \right) x_{i,0}$$

Minimization of Squared Loss

$$L = \sum_{i=1}^m \left(y_i - \sum_{j=0}^n w_j x_{i,j} \right)^2$$

$$\frac{\partial L}{\partial w_0} = \sum_{i=1}^m -2 \left(y_i - \sum_{j=0}^n w_j x_{i,j} \right) x_{i,0}$$

$$\frac{\partial L}{\partial w_1} = \sum_{i=1}^m -2 \left(y_i - \sum_{j=0}^n w_j x_{i,j} \right) x_{i,1}$$

Minimization of Squared Loss

$$L = \sum_{i=1}^m \left(y_i - \sum_{j=0}^n w_j x_{i,j} \right)^2$$

$$\frac{\partial L}{\partial w_0} = \sum_{i=1}^m -2 \left(y_i - \sum_{j=0}^n w_j x_{i,j} \right) x_{i,0}$$

$$\frac{\partial L}{\partial w_1} = \sum_{i=1}^m -2 \left(y_i - \sum_{j=0}^n w_j x_{i,j} \right) x_{i,1}$$

$\vdots$

$$\frac{\partial L}{\partial w_n} = \sum_{i=1}^m -2 \left(y_i - \sum_{j=0}^n w_j x_{i,j} \right) x_{i,n}$$

Minimization of Squared Loss

$$\frac{\partial L}{\partial w_0} = \sum_{i=1}^m -2 \left(y_i - \sum_{j=0}^n w_j x_{i,j} \right) x_{i,0}$$

$$\frac{\partial L}{\partial w_1} = \sum_{i=1}^m -2 \left(y_i - \sum_{j=0}^n w_j x_{i,j} \right) x_{i,1}$$

$\vdots$

$$\frac{\partial L}{\partial w_n} = \sum_{i=1}^m -2 \left(y_i - \sum_{j=0}^n w_j x_{i,j} \right) x_{i,n}$$

Minimization of Squared Loss

$$0 = \sum_{i=1}^m -2 \left(y_i - \sum_{j=0}^n w_j x_{i,j} \right) x_{i,0}$$

$$0 = \sum_{i=1}^m -2 \left(y_i - \sum_{j=0}^n w_j x_{i,j} \right) x_{i,1}$$

$\vdots$

$$0 = \sum_{i=1}^m -2 \left(y_i - \sum_{j=0}^n w_j x_{i,j} \right) x_{i,n}$$

Minimization of Squared Loss

$$\begin{aligned} 0 &= \sum_{i=1}^m \left(y_i - \sum_{j=0}^n w_j x_{i,j} \right) x_{i,0} \\ 0 &= \sum_{i=1}^m \left(y_i - \sum_{j=0}^n w_j x_{i,j} \right) x_{i,1} \\ &\vdots \\ 0 &= \sum_{i=1}^m \left(y_i - \sum_{j=0}^n w_j x_{i,j} \right) x_{i,n} \end{aligned}$$

Minimization of Squared Loss

$$\begin{aligned} 0 &= \sum_{i=1}^m \left(y_i x_{i,0} - \sum_{j=0}^n w_j x_{i,j} x_{i,0} \right) \\ 0 &= \sum_{i=1}^m \left(y_i x_{i,1} - \sum_{j=0}^n w_j x_{i,j} x_{i,1} \right) \\ &\vdots \\ 0 &= \sum_{i=1}^m \left(y_i x_{i,n} - \sum_{j=0}^n w_j x_{i,j} x_{i,n} \right) \end{aligned}$$

Minimization of Squared Loss

$$\begin{aligned}0 &= \sum_{i=1}^m y_i x_{i,0} - \sum_{i=1}^m \sum_{j=0}^n w_j x_{i,j} x_{i,0} \\0 &= \sum_{i=1}^m y_i x_{i,1} - \sum_{i=1}^m \sum_{j=0}^n w_j x_{i,j} x_{i,1} \\&\vdots \\0 &= \sum_{i=1}^m y_i x_{i,n} - \sum_{i=1}^m \sum_{j=0}^n w_j x_{i,j} x_{i,n}\end{aligned}$$

Minimization of Squared Loss

$$\begin{aligned}0 &= \sum_{i=1}^m y_i x_{i,0} - \sum_{j=0}^n \sum_{i=1}^m w_j x_{i,j} x_{i,0} \\0 &= \sum_{i=1}^m y_i x_{i,1} - \sum_{j=0}^n \sum_{i=1}^m w_j x_{i,j} x_{i,1} \\&\vdots \\0 &= \sum_{i=1}^m y_i x_{i,n} - \sum_{j=0}^n \sum_{i=1}^m w_j x_{i,j} x_{i,n}\end{aligned}$$

Minimization of Squared Loss

$$\begin{aligned}0 &= \sum_{i=1}^m y_i x_{i,0} - \sum_{j=0}^n \left(\sum_{i=1}^m x_{i,0} x_{i,j} \right) w_j \\0 &= \sum_{i=1}^m y_i x_{i,1} - \sum_{j=0}^n \left(\sum_{i=1}^m x_{i,1} x_{i,j} \right) w_j \\&\vdots \\0 &= \sum_{i=1}^m y_i x_{i,n} - \sum_{j=0}^n \left(\sum_{i=1}^m x_{i,n} x_{i,j} \right) w_j\end{aligned}$$

Minimization of Squared Loss

$$\begin{aligned}0 &= \sum_{i=1}^m y_i x_{i,0} - \sum_{j=0}^n \left(\sum_{i=1}^m x_{i,0} x_{i,j} \right) w_j \\0 &= \sum_{i=1}^m y_i x_{i,1} - \sum_{j=0}^n \left(\sum_{i=1}^m x_{i,1} x_{i,j} \right) w_j \\&\vdots \\0 &= \sum_{i=1}^m y_i x_{i,n} - \sum_{j=0}^n \left(\sum_{i=1}^m x_{i,n} x_{i,j} \right) w_j\end{aligned}$$

where

$$\begin{aligned}A_{k,l} &= \sum_{i=1}^m x_{i,k} x_{i,l} \\c_k &= \sum_{i=1}^m y_i x_{i,k}\end{aligned}$$

Minimization of Squared Loss

$$0 = c_0 - \sum_{j=0}^n A_{0,j} w_j$$

$$0 = c_1 - \sum_{j=0}^n A_{1,j} w_j$$

$$\vdots$$

$$0 = c_n - \sum_{j=0}^n A_{n,j} w_j$$

where

$$A_{k,l} = \sum_{i=1}^m x_{i,k} x_{i,l}$$

$$c_k = \sum_{i=1}^m y_i x_{i,k}$$

Minimization of Squared Loss

$$0 = c_0 - \sum_{j=0}^n A_{0,j} w_j$$

$$0 = c_1 - \sum_{j=0}^n A_{1,j} w_j$$

$$\vdots$$

$$0 = c_n - \sum_{j=0}^n A_{n,j} w_j$$

where

$$A_{k,l} = \sum_{i=1}^m x_{i,k} x_{i,l}$$

$$c_k = \sum_{i=1}^m y_i x_{i,k}$$

$$\mathbf{A} = \mathbf{X}^T \mathbf{X}$$

$$\vec{c} = \mathbf{X}^T \mathbf{Y}$$

Minimization of Squared Loss

$$0 = c_0 - \sum_{j=0}^n A_{0,j} w_j$$

$$0 = c_1 - \sum_{j=0}^n A_{1,j} w_j$$

$$\vdots$$

$$0 = c_n - \sum_{j=0}^n A_{n,j} w_j$$

$$\vec{0} = \vec{c} - A\vec{w}$$

where

$$A_{k,l} = \sum_{i=1}^m x_{i,k} x_{i,l}$$

$$A = X^T X$$

$$c_k = \sum_{i=1}^m y_i x_{i,k}$$

$$\vec{c} = X^T Y$$

Minimization of Squared Loss

$$c_0 = \sum_{j=0}^n A_{0,j} w_j$$

$$c_1 = \sum_{j=0}^n A_{1,j} w_j$$

$$\vdots$$

$$c_n = \sum_{j=0}^n A_{n,j} w_j$$

$$\vec{c} = \mathbf{A} \vec{w}$$

where

$$A_{k,l} = \sum_{i=1}^m x_{i,k} x_{i,l}$$

$$\mathbf{A} = \mathbf{X}^\top \mathbf{X}$$

$$c_k = \sum_{i=1}^m y_i x_{i,k}$$

$$\vec{c} = \mathbf{X}^\top \mathbf{Y}$$

Minimization of Squared Loss

$$c_0 = \sum_{j=0}^n A_{0,j} w_j$$

$$c_1 = \sum_{j=0}^n A_{1,j} w_j$$

$$\vdots$$

$$c_n = \sum_{j=0}^n A_{n,j} w_j$$

$$\boxed{\vec{w} = \mathbf{A}^{-1} \vec{c}}$$

where

$$A_{k,l} = \sum_{i=1}^m x_{i,k} x_{i,l}$$

$$c_k = \sum_{i=1}^m y_i x_{i,k}$$

$$\mathbf{A} = \mathbf{X}^T \mathbf{X}$$

$$\vec{c} = \mathbf{X}^T \mathbf{Y}$$

Minimization of Squared Loss, reprise

$$L = \sum_{i=1}^m \left(y_i - \sum_{j=0}^n w_j x_{i,j} \right)^2$$

Minimization of Squared Loss, reprise

$$L = \sum_{i=1}^m \left(y_i - \sum_{j=0}^n w_j x_{i,j} \right)^2$$

$$\begin{bmatrix} \sum_{j=0}^n x_{1,j} w_j \\ \sum_{j=0}^n x_{2,j} w_j \\ \vdots \\ \sum_{j=0}^n x_{m,j} w_j \end{bmatrix} = \begin{bmatrix} \text{---} x_{1,\cdot} \text{---} \\ \text{---} x_{2,\cdot} \text{---} \\ \vdots \\ \text{---} x_{m,\cdot} \text{---} \end{bmatrix} \begin{bmatrix} w_0 \\ w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix} = \mathbf{X} \vec{w}$$

Minimization of Squared Loss, reprise

$$L = \sum_{i=1}^m \left(y_i - \sum_{j=0}^n w_j x_{i,j} \right)^2$$

$$\begin{bmatrix} y_1 - \sum_{j=0}^n x_{1,j} w_j \\ y_2 - \sum_{j=0}^n x_{2,j} w_j \\ \vdots \\ y_m - \sum_{j=0}^n x_{m,j} w_j \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix} - \begin{bmatrix} \text{---} x_{1,\text{---}} \text{---} \\ \text{---} x_{2,\text{---}} \text{---} \\ \vdots \\ \text{---} x_{m,\text{---}} \text{---} \end{bmatrix} \begin{bmatrix} w_0 \\ w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix} = \mathbf{Y} - \mathbf{X} \vec{w}$$

Minimization of Squared Loss, reprise

$$L = \sum_{i=1}^m \left(y_i - \sum_{j=0}^n w_j x_{i,j} \right)^2$$
$$L = (\mathbf{Y} - \mathbf{X}\vec{w})^\top (\mathbf{Y} - \mathbf{X}\vec{w})$$

$$\begin{bmatrix} y_1 - \sum_{j=0}^n x_{1,j} w_j \\ y_2 - \sum_{j=0}^n x_{2,j} w_j \\ \vdots \\ y_m - \sum_{j=0}^n x_{m,j} w_j \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix} - \begin{bmatrix} \text{---} x_{1,\text{---}} \text{---} \\ \text{---} x_{2,\text{---}} \text{---} \\ \vdots \\ \text{---} x_{m,\text{---}} \text{---} \end{bmatrix} \begin{bmatrix} w_0 \\ w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix} = \mathbf{Y} - \mathbf{X}\vec{w}$$

Minimization of Squared Loss, reprise

$$L = (\mathbf{Y} - \mathbf{X}\vec{w})^\top (\mathbf{Y} - \mathbf{X}\vec{w})$$

Minimization of Squared Loss, reprise

$$L = (\mathbf{Y} - \mathbf{X}\vec{w})^\top (\mathbf{Y} - \mathbf{X}\vec{w})$$
$$\frac{\partial L}{\partial \vec{w}} =$$

Minimization of Squared Loss, reprise

$$L = (\mathbf{Y} - \mathbf{X}\vec{w})^\top (\mathbf{Y} - \mathbf{X}\vec{w})$$
$$\frac{\partial L}{\partial \vec{w}} = \nabla_{\vec{w}} L$$

Minimization of Squared Loss, reprise

$$L = (\mathbf{Y} - \mathbf{X}\vec{w})^\top (\mathbf{Y} - \mathbf{X}\vec{w})$$
$$\frac{\partial L}{\partial \vec{w}} = \nabla_{\vec{w}} L = -2(\mathbf{Y} - \mathbf{X}\vec{w})^\top \mathbf{X}$$

Minimization of Squared Loss, reprise

$$\begin{aligned} L &= (\mathbf{Y} - \mathbf{X}\vec{w})^\top (\mathbf{Y} - \mathbf{X}\vec{w}) \\ \frac{\partial L}{\partial \vec{w}} &= \nabla_{\vec{w}} L = -2(\mathbf{Y} - \mathbf{X}\vec{w})^\top \mathbf{X} \\ &= -2(\mathbf{Y}^\top \mathbf{X} - \vec{w}^\top \mathbf{X}^\top \mathbf{X}) \end{aligned}$$

Minimization of Squared Loss, reprise

$$\begin{aligned}L &= (\mathbf{Y} - \mathbf{X}\vec{w})^\top (\mathbf{Y} - \mathbf{X}\vec{w}) \\ \frac{\partial L}{\partial \vec{w}} &= \nabla_{\vec{w}} L = -2(\mathbf{Y} - \mathbf{X}\vec{w})^\top \mathbf{X} \\ &= -2(\mathbf{Y}^\top \mathbf{X} - \vec{w}^\top \mathbf{X}^\top \mathbf{X}) \\ \vec{0}^\top &= -2(\mathbf{Y}^\top \mathbf{X} - \vec{w}^\top \mathbf{X}^\top \mathbf{X})\end{aligned}$$

Minimization of Squared Loss, reprise

$$\begin{aligned}L &= (\mathbf{Y} - \mathbf{X}\vec{w})^\top (\mathbf{Y} - \mathbf{X}\vec{w}) \\ \frac{\partial L}{\partial \vec{w}} &= \nabla_{\vec{w}} L = -2(\mathbf{Y} - \mathbf{X}\vec{w})^\top \mathbf{X} \\ &= -2(\mathbf{Y}^\top \mathbf{X} - \vec{w}^\top \mathbf{X}^\top \mathbf{X}) \\ \vec{0}^\top &= -2(\mathbf{Y}^\top \mathbf{X} - \vec{w}^\top \mathbf{X}^\top \mathbf{X}) \\ \vec{0} &= \mathbf{X}^\top \mathbf{Y} - \mathbf{X}^\top \mathbf{X} \vec{w}\end{aligned}$$

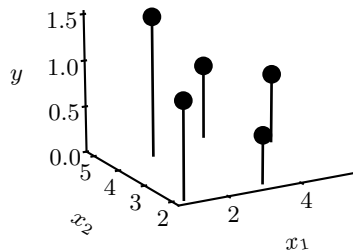
Minimization of Squared Loss, reprise

$$\begin{aligned}L &= (\mathbf{Y} - \mathbf{X}\vec{w})^\top (\mathbf{Y} - \mathbf{X}\vec{w}) \\ \frac{\partial L}{\partial \vec{w}} &= \nabla_{\vec{w}} L = -2(\mathbf{Y} - \mathbf{X}\vec{w})^\top \mathbf{X} \\ &= -2(\mathbf{Y}^\top \mathbf{X} - \vec{w}^\top \mathbf{X}^\top \mathbf{X}) \\ \vec{0}^\top &= -2(\mathbf{Y}^\top \mathbf{X} - \vec{w}^\top \mathbf{X}^\top \mathbf{X}) \\ \vec{0} &= \mathbf{X}^\top \mathbf{Y} - \mathbf{X}^\top \mathbf{X} \vec{w} \\ \mathbf{X}^\top \mathbf{X} \vec{w} &= \mathbf{X}^\top \mathbf{Y}\end{aligned}$$

Minimization of Squared Loss, reprise

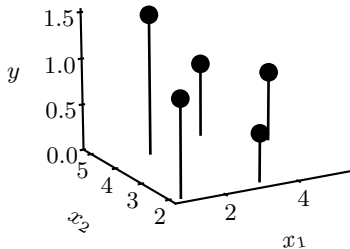
$$\begin{aligned}L &= (\mathbf{Y} - \mathbf{X}\vec{w})^\top (\mathbf{Y} - \mathbf{X}\vec{w}) \\ \frac{\partial L}{\partial \vec{w}} &= \nabla_{\vec{w}} L = -2(\mathbf{Y} - \mathbf{X}\vec{w})^\top \mathbf{X} \\ &= -2(\mathbf{Y}^\top \mathbf{X} - \vec{w}^\top \mathbf{X}^\top \mathbf{X}) \\ \vec{0}^\top &= -2(\mathbf{Y}^\top \mathbf{X} - \vec{w}^\top \mathbf{X}^\top \mathbf{X}) \\ \vec{0} &= \mathbf{X}^\top \mathbf{Y} - \mathbf{X}^\top \mathbf{X} \vec{w} \\ \mathbf{X}^\top \mathbf{X} \vec{w} &= \mathbf{X}^\top \mathbf{Y} \\ \vec{w} &= (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}\end{aligned}$$

Least-Squares, Example



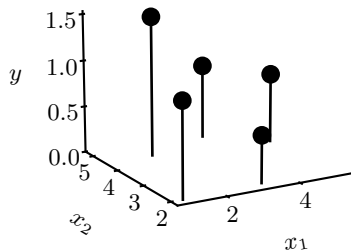
$$\mathbf{X} = \begin{bmatrix} 1.0 & 2.0 \\ 2.0 & 4.5 \\ 3.2 & 2.1 \\ 3.9 & 5.1 \\ 5.2 & 4.2 \end{bmatrix} \quad \mathbf{Y} = \begin{bmatrix} 1.0 \\ 1.5 \\ 0.5 \\ 0.8 \\ 0.75 \end{bmatrix}$$

Least-Squares, Example



$$\mathbf{X} = \begin{bmatrix} 1.0 & 1.0 & 2.0 \\ 1.0 & 2.0 & 4.5 \\ 1.0 & 3.2 & 2.1 \\ 1.0 & 3.9 & 5.1 \\ 1.0 & 5.2 & 4.2 \end{bmatrix} \quad \mathbf{Y} = \begin{bmatrix} 1.0 \\ 1.5 \\ 0.5 \\ 0.8 \\ 0.75 \end{bmatrix}$$

Least-Squares, Example

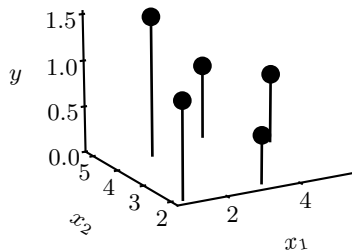


$$\mathbf{X} = \begin{bmatrix} 1.0 & 1.0 & 2.0 \\ 1.0 & 2.0 & 4.5 \\ 1.0 & 3.2 & 2.1 \\ 1.0 & 3.9 & 5.1 \\ 1.0 & 5.2 & 4.2 \end{bmatrix} \quad \mathbf{Y} = \begin{bmatrix} 1.0 \\ 1.5 \\ 0.5 \\ 0.8 \\ 0.75 \end{bmatrix}$$

$$\mathbf{A} = \mathbf{X}^T \mathbf{X} = \begin{bmatrix} 5.0 & 15.3 & 17.9 \\ 15.3 & 57.49 & 59.45 \\ 17.9 & 59.45 & 72.31 \end{bmatrix}$$

$$\vec{c} = \mathbf{X}^T \mathbf{Y} = \begin{bmatrix} 4.55 \\ 12.62 \\ 17.03 \end{bmatrix}$$

Least-Squares, Example



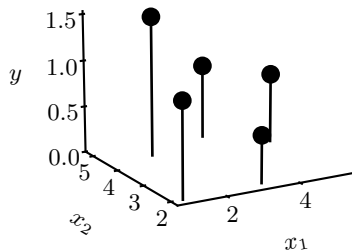
$$\mathbf{X} = \begin{bmatrix} 1.0 & 1.0 & 2.0 \\ 1.0 & 2.0 & 4.5 \\ 1.0 & 3.2 & 2.1 \\ 1.0 & 3.9 & 5.1 \\ 1.0 & 5.2 & 4.2 \end{bmatrix} \quad \mathbf{Y} = \begin{bmatrix} 1.0 \\ 1.5 \\ 0.5 \\ 0.8 \\ 0.75 \end{bmatrix}$$

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Least-Squares, Example



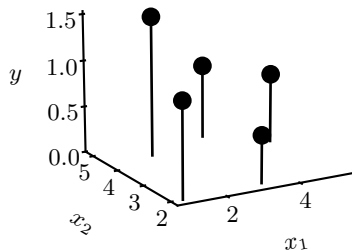
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$$\vec{w} = \begin{bmatrix} 5.0 & 15.3 & 17.9 \\ 15.3 & 57.49 & 59.45 \\ 17.9 & 59.45 & 72.31 \end{bmatrix}^{-1} \begin{bmatrix} 4.55 \\ 12.62 \\ 17.03 \end{bmatrix}$$

Least-Squares, Example



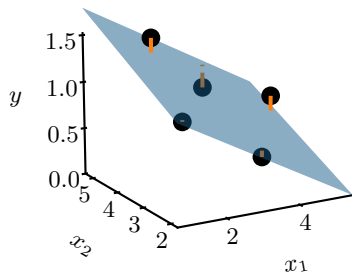
$$\mathbf{X} = \begin{bmatrix} 1.0 & 1.0 & 2.0 \\ 1.0 & 2.0 & 4.5 \\ 1.0 & 3.2 & 2.1 \\ 1.0 & 3.9 & 5.1 \\ 1.0 & 5.2 & 4.2 \end{bmatrix} \quad \mathbf{Y} = \begin{bmatrix} 1.0 \\ 1.5 \\ 0.5 \\ 0.8 \\ 0.75 \end{bmatrix}$$

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$$\vec{c} = \mathbf{X}^T \mathbf{Y} = \begin{bmatrix} 4.55 \\ 12.62 \\ 17.03 \end{bmatrix}$$

$$\vec{w} = \begin{bmatrix} 0.808 \\ -0.215 \\ 0.212 \end{bmatrix}$$

Least-Squares, Example



$$\mathbf{X} = \begin{bmatrix} 1.0 & 1.0 & 2.0 \\ 1.0 & 2.0 & 4.5 \\ 1.0 & 3.2 & 2.1 \\ 1.0 & 3.9 & 5.1 \\ 1.0 & 5.2 & 4.2 \end{bmatrix} \quad \mathbf{Y} = \begin{bmatrix} 1.0 \\ 1.5 \\ 0.5 \\ 0.8 \\ 0.75 \end{bmatrix}$$

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